

AI Opportunities During Product Concept Phase - ABC Homes

Completed AI process analysis and Balanced Scorecard business case

ABC Homes - AI Opportunities and Balanced Scorecard Business Case

Task 1: AI Opportunities in the Current Six-Step Process

Review the ABC Homes household cleaners scenario. Explain how a generative AI solution can enhance product research, market research, formulation, testing, evaluation, and production/distribution for less toxic cleaners.

Process Step	Generative AI Opportunity
Product research	Summarize chemical safety literature, compare ingredient toxicity profiles, identify greener substitutes, and generate hypotheses for safer formulations.
Market research	Analyze competitor claims, customer reviews, product lifecycle signals, pricing, and unmet customer needs for environmentally safer cleaners.
Formulation	Suggest candidate formulations based on desired cleaning performance, safety constraints, cost targets, ingredient compatibility, and sustainability goals.
Testing	Create testing protocols, predict likely failure points, organize lab results, and recommend refinement paths based on test data.
Evaluation	Draft feasibility summaries, risk assessments, business case inputs, and go/no-go recommendations using structured criteria.
Production and distribution	Support supplier analysis, packaging options, demand forecasts, quality documentation, and launch-readiness checklists.

Task 2: Balanced Scorecard Business Case

Analyze how a generative AI solution for ABC Homes impacts Finance, Process, Employees, and Customers under the Balanced Scorecard framework.

Balanced Scorecard Area	Business Case Impact
Finance	Potential cost avoidance from fewer failed formulation attempts, reduced waste, faster research cycles, better supplier comparison, and improved probability of launching competitive products. Revenue may improve if safer cleaners win share from environmentally conscious customers.
Process	Shortens research and formulation cycles, creates repeatable workflows, improves documentation quality, accelerates competitive analysis, and helps teams move from trial-and-error to data-informed experimentation.
Employees	Reduces repetitive research and documentation workload, helps chemists and marketers collaborate using shared evidence, and frees staff to focus on judgment, safety validation, and strategy.
Customers	Supports safer, effective, reasonably priced cleaners with clearer claims, better quality, and stronger alignment with environmental and health expectations.

Conclusion

Generative AI should be treated as a decision-support capability, not a replacement for scientific testing or regulatory review. The strongest business case is faster learning with better evidence, lower waste, and stronger customer alignment.